Why These Simple Laws, Forward Time, Many-Worlds, and Superdeterminism

Gary Abraham Bernstein

Independent Researcher ORCID: https://orcid.org/0009-0009-1761-2867

Abstract

As reality is mathematical structure, derived from the pattern-randomness dichotomy (Bernstein, 2026c), this paper derives the physical consequences. Under algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000), the measure over all consistent structures is P(S) proportional to 2^(-K(S)), where K is Kolmogorov complexity. Each additional bit of specification halves a structure's measure. From this single principle, convergent results follow: (1) simple laws dominate the measure, explaining why the laws of physics fit on a page; (2) the Principle of Least Action is a selection effect: variational specifications have lower K; (3) cosmological fine-tuning dissolves: the constants are among the simplest compatible with observers; (4) the arrow of time follows: forward-described structures have lower K, grounded in the irreversibility of computation; (5) Many-Worlds has the lowest specification cost among quantum interpretations: collapse accumulates K per measurement, MWI does not; (6) superdeterminism survives the fine-tuning objection: specificity is not complexity; (7) string theory is exponentially disfavored: landscape-selection requires 3,000+ bits vs a few hundred for the Standard Model; (8) the hard problem of consciousness assumes its conclusion: any explanation must itself be structured; (9) the Principle of Least Action, the arrow of time, and emergence are three instances of one structural fact: many-to-one mappings; (10) extra dimensions emerge free: the metric bundle (14D over 4D) and its infinite jet bundle tower are mathematically necessary at zero additional K. Under algorithmic probability, elegance is short description, and string theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds. The position differs from Tegmark (2014) in deriving the ensemble rather than positing it, and from Schmidhuber (2000) in grounding why the ensemble exists.

Keywords: mathematical universe, algorithmic probability, Kolmogorov complexity, many-worlds, superdeterminism, arrow of time, fine-tuning, string theory, consciousness, mathematical monism

---

1. Introduction

Reality must be mathematical structure. The following argument aims to derive this conclusion, not merely hypothesize it. The derivation follows from an exhaustive dichotomy: all existence is either patterned or not-patterned (random).

Patterns are regularities expressible through mathematical relations. Randomness is characterized via probability distributions and algorithmic incompressibility (Kolmogorov complexity). Both categories are inherently mathematical. There is no third option; even structurelessness is structure, since complete random is mathematically defined. This exhaustive dichotomy, developed in Section 2, establishes that whatever exists, whatever has determinate features distinguishing it from nothing, must be mathematical structure.

From this foundation, striking consequences follow. If existence is identical to mathematical consistency, all self-consistent structures exist. Since any filter is itself a consistent structure, all filters exist, including the null filter. The Level IV multiverse emerges as deductive consequence (Section 3) rather than cosmological conjecture. This resolves the grounding problem facing Tegmark's (2008, 2014) Mathematical Universe Hypothesis: he posited mathematical existence; the present argument derives it.

But the multiverse raises a new puzzle: the measure problem. Among all structures, which should observers expect to inhabit? Without a measure, predictions are impossible. Certain naive measures yield catastrophe (Boltzmann brains dominating). Section 4 proposes algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000): structures are weighted by algorithmic probability, P(S)   2^(−K(S)), where K(S) is Kolmogorov complexity. Algorithmic probability is uniquely motivated by information-theoretic considerations. Moreover, since reality *is* structure, the measure is not chosen but discovered: algorithmic probability is the natural measure over structure-space, not an external imposition.

Section 4 derives consequences and testable predictions, including a novel explanation for why physical laws take variational form: algorithmic probability selects for universes whose physics follows the Principle of Least Action, because variational specifications have lower generative complexity. Section 5 examines implications for quantum mechanics. Section 7 responds to objections.

The position developed here (call it *mathematical monism*, or *Matheism*) completes Ontic Structural Realism (OSR) as defended by Ladyman and Ross (2007) and French (2014, 2022). Standard OSR describes structure as reality's sole constituent but does not explain *why* structure exists. The pattern-randomness dichotomy closes this gap: mathematical consistency is self-necessitating; consistent structures cannot fail to exist, just as 2+2 cannot fail to equal 4. The philosophical derivation is given in Bernstein (2026c); the present paper develops the physical consequences.

---

2. Foundation

A companion paper (Bernstein, 2026c) establishes the full argument; what follows is the core.

Anything that exists must have determinate features, or it is indistinguishable from nothing. Determinate features are either patterned (regularities governed by rules) or random (algorithmically incompressible). Both are mathematical. The pattern-randomness dichotomy is exhaustive: no coherent third category exists. Only consistent mathematics forms structure: inconsistent statements can be labeled but not constructed.

Physical reality and its isomorphic mathematical structure share all relational properties. By the Identity of Indiscernibles, they are identical. The map is the territory. Any proposed non-mathematical substrate is explanatorily idle: it makes no observable difference and, by the dichotomy, is itself structure. Physics has progressively confirmed this: quantum field theory contains no objects with intrinsic properties independent of their relational role. The Standard Model is purely relational.

A transcendental constraint: observers are logically constrained to patterned realities. Observation requires memory, perception, and stable structure, all of which presuppose pattern. "Observer in brute randomness" is incoherent, like "four-sided triangle."

Any two things that interact must share structure for causal commerce. Therefore any interacting dualism collapses into mathematical monism. The principle extends to God: any deity with a nature has structure; any deity without a nature is indistinguishable from nothing.

All consistent structures exist. Any filter selecting among them is itself a consistent structure. All filters exist, including the null filter that excludes nothing. The Level IV multiverse follows as deductive consequence, not cosmological conjecture. Unlike Tegmark (2014), who posits this, the present argument derives it. The Principle of Sufficient Reason is satisfied: explanatory chains terminate at mathematical necessity. The physical/abstract distinction dissolves: "physical" names structure experienced from within; "abstract" names structure viewed from outside. Qualia are what mathematics feels like from inside.

The consequences for philosophy are immediate. The hard problem of consciousness is circular reasoning. It assumes sentience cannot be structure at the same time as the dichotomy derives that substance can only be structure. Sentience is experiencing math from the inside. Any proposed bridge across the sentience-structure gap can only be more language, more structure, more mathematics, so it can never cross the gap it presupposes.

All dualisms collapse into one mathematical monism, like yin and yang interacting in one circle. Interaction requires shared structure, and shared structure is the monism. Panpsychism inherits the interaction problem it claims to solve and cannot explain why minds find math everywhere. Mathematical monism explains both without incoherence.

The supernatural is a square circle: any deity with a nature has structure, any deity without a nature is nothing. Libertarian free will is a square circle:

"free" requires randomness, "will" requires determinism, and the conjunction is incoherent. Fundamental dualism of any kind is a square circle. The pattern-randomness dichotomy leaves no room for a non-mathematical category. What remains is one mathematical reality, uncreated, necessary, and indestructible. The physics follows.

3. The Measure Problem and algorithmic probability

3.1 The Measure Problem

The Level IV multiverse resolves fine-tuning in principle but raises a new question: among all consistent structures, which should we expect to observe? Without a measure over structures, we cannot make predictions.

One might object that the question is meaningless: all structures exist equally, so there is no 'probability' of being in one rather than another. But this proves too much. If all structures are equally weighted, we should expect to find ourselves in a 'typical' structure. What counts as typical? Without a measure, we cannot say, but some measures yield absurd conclusions (e.g., under certain measures, almost all observers are Boltzmann brains with false memories).

A crucial insight transforms this problem: Since reality *is* structure (derived via the dichotomy), the measure is intrinsic. We identify the measure structure-space already has. Algorithmic probability is the natural measure over computational/structural space, arising from the combinatorics of description length. This transforms the measure problem from "which measure should we choose?" to "what measure does structure-space have?" The answer is algorithmic probability, for the same reason the natural measure over integers weights small numbers more heavily: there are more ways to specify them.

I propose:

Algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000): The measure of a mathematical structure S is governed by its algorithmic probability:

$$P(S) propto 2^{-K(S)}$$

where K(S) is the Kolmogorov complexity of S, that is, the length of the shortest program that generates the structure's rules.

This approach builds on Schmidhuber's (2000, 2002) algorithmic theories of everything, which first rigorously proposed that universes are weighted by algorithmic probability. Schmidhuber further refined this with the Speed Prior, incorporating computation time alongside program length. The contribution here is not scope but grounding: deriving *why* the ensemble exists (via the null filter argument) and *why* algorithmic probability governs it, rather than positing both as starting assumptions. Positing an ensemble is assertion; deriving it closes the objection “why this ensemble rather than another?” The null filter argument shows the ensemble is not one hypothesis among alternatives but the unique non-arbitrary solution. Similarly, positing algorithmic probability as the measure invites “why this measure?”; grounding it in the combinatorics of program length shows it was always there. Tegmark’s (2008) Mathematical Universe Hypothesis asserts that all mathematical structures exist but does not derive it; Schmidhuber’s framework provides the measure but posits the ensemble. The present work derives the ensemble a priori, then applies algorithmic probability as its natural measure.

In practice, non-computable structures have undefined K, so algorithmic probability assigns them no determinate measure; only computable structures contribute to predictions. The inclusion of all consistent structures is for logical completeness, as the pattern-randomness dichotomy does not exclude them a priori. More precisely: non-computable structures require oracle machines, which depend on oracles encoding infinite brute facts (e.g., which programs halt). If there were a pattern, the oracle would be computable. So oracles are maximally random, with $K = \infty$, yielding measure $2\hat{}(-\infty) = 0$. Non-computable structures exist but have measure zero. This derives Schmidhuber’s restriction to computable structures from the measure itself, rather than imposing it by fiat.

Note that simple, finitely specifiable laws can generate non-computable phenomena as outputs (e.g., undecidable spectral gaps, chaotic systems requiring infinite precision). The laws have low K; some consequences may not. The measure weighs structures by the complexity of their generating rules, not their outputs. Theories with continuous dynamics but finite specification (such as string theory) remain consistent with algorithmic probability; the constraint is finite specifiability, which discrete approaches satisfy manifestly.

Critical distinction: K(S) measures *generative* complexity (the shortest program specifying the laws or rules), not *instance* complexity of specific states. A universe with simple laws (low generative complexity) can produce configurations of enormous detail, even infinite states. The Mandelbrot set has low K(S) (its definition is ~50 characters) but infinite fractal complexity.

3.2 Three Types of Probability

Three distinct concepts travel under ‘probability’ here; conflating them breeds confusion.

Quantum probability concerns credences within a branch. If Many-Worlds is correct, all measurement outcomes occur; the Born rule (that quantum probabilities follow the square of the wave amplitude) governs rational expectation about which branch contains one’s continuation. This is epistemic: indexical uncertainty, not ontological indeterminacy.

Measure-theoretic weight concerns the distribution over mathematical structures. Simpler structures have higher algorithmic probability. This is not about which structures ‘get created’ (all exist timelessly) but about their relative weight in structure-space, analogous to how shorter programs outnumber longer ones combinatorially.

Anthropic constraint concerns which structures can contain observers. Combined with measure, this yields predictions: observers should expect to find themselves in high-measure, observer-supporting structures.

These three probabilities are related but not identical. Quantum probability operates within a structure; measure-theoretic weight operates over structures; anthropic constraint filters the structures relevant to prediction.

3.3 Why Algorithmic Probability?

Four converging justifications support algorithmic probability:

Logical-space justification: In hypothesis space, simpler hypotheses make fewer arbitrary commitments and therefore cover more possibilities. A hypothesis specifying 10 parameters is compatible with fewer worlds than one specifying 5 parameters. Simpler structures are less constrained, hence ‘larger’ in logical space.

Measure-theoretic justification: In a prefix-free coding, each program of length k occupies a fraction 2^(-k) of program-space. A 10-bit program occupies fraction 2^(-10). A 1000-bit program occupies fraction 2^(-1000). The short program has 2^990 times higher measure, not because short programs are more numerous (they are not: there are more long programs than short ones) but because each short program claims a larger fraction of the space. If we imagine a universal Turing machine fed random bits, the probability of hitting any specific short program is exponentially higher than hitting any specific long one. This is Solomonoff’s (1964) foundation for inductive inference; algorithmic probability extends it from epistemology to ontology. A “program” here is a finite description, not a process requiring execution. The structures exist as mathematics. The measure is over specifications, not computations. Schmidhuber (1997, 2000, 2002) developed this into algorithmic theories of everything; the present argument provides the a priori derivation for why the ensemble exists.

Coherence justification: Algorithmic probability makes our universe not miraculous. Physical laws take simple forms (the Standard Model Lagrangian, the single function encoding all particle physics as kinetic minus potential energy, fits on a t-shirt; general relativity derives from a one-line action principle). If all structures existed with equal measure, simple laws would be vanishingly unlikely among observer-supporting structures. Algorithmic probability explains why simplicity prevails.

Invariance justification: Algorithmic probability is the only measure invariant under computable transformations up to multiplicative constants. Any alternative measure either fails invariance (making it coordinate-dependent) or reduces to algorithmic probability. This uniqueness makes algorithmic probability not one arbitrary choice among many, but the canonical choice dictated by the structure of computation itself.

A potential circularity deserves address: algorithmic probability is defined relative to Turing machines, but Turing machines are themselves mathematical structures. Does this make algorithmic probability viciously circular, defining the measure over all structures in terms of one particular structure?

The response: Turing computability is not an arbitrary structure but a *universal* one. The Church-Turing thesis establishes that all reasonable notions of effective computation converge on the same class of functions. Different models (Turing machines, lambda calculus, recursive functions, cellular automata) yield identical computability classes. This universality is itself a mathematical theorem, not an empirical accident. Algorithmic probability inherits this universality: the measure is invariant (up to multiplicative constants) across different universal machines. We are not privileging one structure among equals; we are identifying a canonical measure that emerges from the structure of computation as such, a structure that every sufficiently rich mathematical framework contains.

Moreover, even if one resists this universality argument, algorithmic probability remains the best available hypothesis. The burden falls on critics: produce an alternative measure with comparable theoretical virtues. None has been offered.

This uniqueness deserves emphasis. The critic asking 'why algorithmic probability rather than some other measure?' faces a burden: *produce* an alternative satisfying comparable desiderata. Lebesgue measure is undefined over the relevant space (no natural uniform distribution over an infinite ensemble with unbounded cardinality). Uniform measure over bounded complexity is arbitrary at the bound. Measures favoring complexity face the Boltzmann brain catastrophe. Algorithmic probability is the only measure with independent mathematical motivation (Solomonoff induction), the only measure avoiding known pathologies, and the only measure explaining the observed success of parsimony in science.

3.4 Occam's Razor as Metaphysical Principle

Algorithmic probability provides a grounding for Occam's Razor. Traditionally, parsimony is treated as methodological preference: we should prefer simpler theories, but without clear justification for why simpler theories are more likely true. Hossenfelder (2018) pressed this point, asking why physics should prefer simplicity at all. Algorithmic probability answers: lower-K structures have exponentially higher measure. The criterion is not beauty but brevity.

Algorithmic probability transforms this. Since simpler structures dominate the measure over all mathematical structures, simpler theories are more likely true because they describe more probable structures. Occam's Razor is not an arbitrary aesthetic preference or pragmatic heuristic; it is a theorem about the structure of reality.

This explains the otherwise mysterious success of parsimony in science. Newton's laws, Maxwell's equations, Einstein's field equations, and the Standard Model Lagrangian all exhibit striking simplicity despite describing enormously complex phenomena. If simplicity had no connection to truth, this track record would be miraculous coincidence. Algorithmic probability explains it: scientists have been unconsciously tracking algorithmic probability, favoring theories that describe high-measure structures.

3.5 Limitations and Open Questions

Algorithmic probability faces technical challenges that deserve acknowledgment:

Non-computability: Kolmogorov complexity K(S) is uncomputable; there is no algorithm that takes a structure and outputs its complexity. This limits our ability to calculate exact measures. However, we can often establish bounds (a structure's complexity is at most the length of any program that generates it) and make comparative judgments.

Machine-dependence: K(S) depends on choice of universal Turing machine. Different machines yield complexities differing by at most a constant (the invariance theorem), but this constant can be large. For practical purposes, we rely on the fact that the exponential relationship in 2^(−K(S)) dominates additive constants: a structure with K = 100 bits has measure ~2^100 times greater than one with K = 200 bits, regardless of machine choice.

Measure underdetermination: Under Lebesgue measure on [0,1], algorithmically random reals have measure 1; under algorithmic probability, computable reals dominate. Both measures are mathematically legitimate. This may represent a fundamental limit: the 'correct' measure over mathematical structures might be metaphysically underdetermined. If so, Algorithmic probability is not the unique solution but one principled choice among several. The underdetermination parallels gauge freedom in physics: multiple mathematically legitimate choices, but some more natural than others. Algorithmic probability is the 'natural gauge', the choice singled out by independent theoretical virtues. I flag this as an open problem rather than a defeater.

Argumentative status: Given the a priori foundation, the Level IV multiverse follows deductively: all consistent structures must exist if existence is mathematical consistency. *Which measure* governs the ensemble is not derivable a priori but is established by inference to best explanation. Algorithmic probability is the best available hypothesis for the measure problem, with theoretical virtues no competitor matches.

Despite these challenges, algorithmic probability provides a principled basis where alternatives offer none.

---

4. Consequences and Predictions

4.1 Fine-Tuning Resolved

The cosmological fine-tuning problem notes that physical constants appear precisely calibrated for complexity and life. Change the fine-structure constant by 1%, and chemistry becomes impossible. Change the cosmological constant (the energy density of empty space, which drives the expansion of the universe) by 10^−120, and galaxies never form.

Standard responses invoke either design (theism) or selection (weak anthropic principle within a multiverse). Algorithmic probability offers a third option: the constants are not ‘tuned’ at all. Among all observer-supporting structures, those with simple laws dominate. Simple laws produce orderly universes where constants take values permitting complexity. Fine-tuning is what simplicity looks like from the inside.

The compressibility of mathematical constants illustrates this. Pi has infinite precision but near-zero Kolmogorov complexity: short formulas generate all its digits from roughly 10 bits. Under algorithmic probability, compressible constants are exponentially more probable. The prediction is that the Standard Model’s 26 parameters follow from a compact generating rule, each potentially as compressible as pi. Whether this is so remains an open empirical question, but the direction is clear: observers are overwhelmingly likely to find themselves in structures with compressible constants.

More precisely: let S_obs be the set of observer-supporting structures. Algorithmic probability predicts we should find ourselves in a structure with low K(S) relative to S_obs. This is consistent with our universe having elegantly simple fundamental laws despite the apparent fine-tuning of constants.

4.2 The Variational Form of Physical Laws

A striking feature of fundamental physics: laws take variational form. Classical mechanics follows from the principle of least action. General relativity derives from the Einstein-Hilbert action. Quantum mechanics admits path-integral formulation. The Standard Model is defined by a Lagrangian. Fermat’s principle governs optics; Hamilton’s principle governs dynamics; the Einstein-Hilbert action governs spacetime geometry.

Why should nature prefer variational principles? Algorithmic probability provides an answer that transforms this from brute fact to structural necessity.

Consider two ways to specify the same physics:

(a) “Systems follow paths extremizing S = L dt”, a compact variational principle

(b) Exhaustive enumeration of all permitted trajectories

Specification (a) has vastly lower generative complexity. A single functional plus a minimization procedure replaces the enumeration of infinitely many trajectories. A universe governed by (a) has lower K(S) than one requiring (b). Under algorithmic probability, variational universes dominate the measure.

Algorithmic probability therefore selects for universes whose physics takes variational form. This is a selection effect across structure-space, not a dynamical consequence within any particular structure. Algorithmic probability predicts that observer-supporting structures should exhibit variational laws, and they do. A deeper result connects PoLA to the computational arrow: both are instances of many-to-one mappings, many paths to one classical survivor, many inputs to one output. The Noether chain (Noether, 1918: observers $\rightarrow$ stability $\rightarrow$ conservation $\rightarrow$ symmetry $\rightarrow$ variational) may force variational physics for any observer-supporting universe. An analogous argument applies to geometry: the metric bundle over a 4D manifold adds near-zero specification and necessarily contains gauge structure, making Weinstein's Geometric Unity the algorithmically favored continuum description of unification.

This unifies three seemingly distinct parsimony principles:

Ontological parsimony: Simpler structures are more probable (algorithmic probability). Structures with lower Kolmogorov complexity dominate the measure over mathematical reality.

Dynamical parsimony: Simpler paths are preferred (Principle of Least Action). Within physics, systems evolve along extremal trajectories rather than arbitrary paths.

Epistemic parsimony: Simpler theories should be believed (Occam's Razor). In scientific inference, we prefer hypotheses with fewer free parameters and simpler formulations.

All three reflect a single underlying truth: algorithmic simplicity is a feature of reality's deep structure. Occam's Razor works because it tracks ontological probability. The Principle of Least Action holds because variational laws minimize generative complexity. The success of parsimony in science is a direct consequence of our inhabiting a structure selected (by measure, not design) for simplicity.

Feynman's path integral formulation (where a particle takes every possible path simultaneously, and the paths cancel except the classical one) provides additional insight. In quantum mechanics, all paths contribute to the propagator, weighted by e^(iS/). The classical path (extremal action) dominates because nearby paths constructively interfere while distant paths destructively cancel. The Principle of Least Action emerges from quantum mechanics as a consequence of phase coherence. This is parsimony at work at yet another level: the interference structure itself favors extremal paths.

One might object that variational formulations are mathematically ubiquitous: the inverse problem of the calculus of variations shows that broad classes of differential equations admit Lagrangian reformulation. If so, PLA would be a mathematical theorem, not evidence for algorithmic probability. But the claim is not that variational *reformulation* is possible; it is that the *fundamental* specification of our physics is natively variational. The Standard Model Lagrangian is not derived from something more basic and then reformulated; it *is* the most compact specification. The laws are born variational, not translated into variational form. This is what algorithmic probability predicts and what we observe.

4.3 The Many-to-One Root

The Principle of Least Action, the arrow of time, and the emergence of smooth physics from discrete rules are three instances of one structural fact: many-to-one mappings. Computation is many-to-one: 1+1=2 is deterministic forward, 2=x+y is not. The inverse does not exist. Feynman's path integral is many-to-one: all quantum paths exist, but paths far from the extremal cancel under phase interference. What survives is the classical trajectory. Coarse-graining is many-to-one: many microstates map to one macrostate. Variational physics is forced by coarse-graining regardless of whether the fundamental rule is variational. A forcing argument completes the picture: observers require stable structures; stable structures require conservation laws; Noether's theorem proves conservation laws require symmetries of a variational formulation.

4.4 Testable Predictions

Algorithmic probability generates falsifiable predictions:

Prediction 1: The final theory of physics should have low Kolmogorov complexity among theories supporting observers. Lower is exponentially more probable. If the final theory requires irreducibly complex mathematical machinery (extensive arbitrary parameters, baroque symmetry groups with no unifying principle, laws that cannot be compactly expressed), this counts as evidence against algorithmic probability.

Specifically: current fundamental physics (Standard Model + General Relativity) has generative complexity roughly ~10^3 bits. If the eventual unified theory has K(S) within an order of magnitude of this, or lower, algorithmic probability is confirmed. If K(S) proves orders of magnitude higher (say, ~10 bits of irreducible specification), algorithmic probability faces serious difficulty.

Prediction 2: Successful physical theories should exhibit 'unreasonable effectiveness': mathematical structures discovered through pure reasoning should describe physical reality with extraordinary precision. This follows from algorithmic probability: if our universe has low K(S), its laws are simple enough to be discovered by finite minds doing mathematics.

Prediction 3: Apparent complexity in physical phenomena should arise from simple underlying rules, not irreducibly complex foundations. Complexity should be *generated* (high instance complexity from low generative complexity), not *fundamental* (high generative complexity at the base level).

Prediction 4: Mathematical structures discovered in one domain should apply successfully to unrelated domains more frequently than a 'mathematics as descriptive tool' hypothesis would predict. The diffusion equation governing heat flow also governs chemical diffusion, population genetics, and option pricing; the logistic map produces identical bifurcations in fluid dynamics, ecology, and cardiac rhythms. This cross-domain resonance reflects shared underlying structure.

Prediction 5: Outstanding physics puzzles will have simple solutions. Dark matter currently lacks a complete explanation. Algorithmic probability predicts its solution will be simple: a low K addition or modification to existing frameworks, not baroque new structures. If dark matter requires dozens of new particle types with arbitrary parameters, algorithmic probability is challenged. The objection "of course solutions will be simple" concedes the point: the expectation of simplicity *is* algorithmic probability, operating as implicit assumption throughout physics. Making it explicit transforms intuition into testable prediction.

Current physics is consistent with all five predictions. The Standard Model, despite its complexity, derives from compact symmetry principles. General relativity follows from a one-line action. Quantum mechanics admits elegant formulations. Cross-domain mathematical transfer is pervasive.

Prediction 6: Theories requiring higher specification complexity than the phenomena they explain are disfavored. String theory's total K is approximately 3,000 to 4,700 bits (including landscape selection, flux specification (fields threading through the extra dimensions), and moduli stabilization (fixing the sizes and shapes of those dimensions)). The Standard Model's generating rule outputs a list of 19 to 26 scalar values. Even without knowing the Standard Model's exact K, the output-type comparison is decisive: generating numbers is structurally simpler than constructing a specific Calabi-Yau geometry (a six-dimensional shape into which the extra dimensions are curled) from 10^500 options. String theory's "explanation" has higher K than what it purports to explain. A full analysis is given in Bernstein (2026g).

Wolfram's Physics Project provides independent convergent evidence: the hypothesis that our universe emerges from extremely simple hypergraph rewriting rules (potentially specifiable in just a few bits) aligns precisely with algorithmic probability's prediction that generative complexity should be minimal. If such a project succeeds in deriving Standard Model physics from minimal rules, algorithmic probability receives strong corroboration. However, verification faces computational barriers: the hypergraph primitives may operate at scales around 10^−93 meters, nearly 10^58 times smaller than the Planck length, making direct simulation intractable. Mathematical derivation rather than brute-force computation may be required to bridge the gap between simple rules and emergent physics.

A precise irony emerges here, though it should not detract from the main point: the simplest possible specification (a few bits of hypergraph rules) may require the most computation to verify, since playing out discrete steps to macroscopic scales demands a universal computer. This inverse relationship between generative simplicity and verification difficulty is not general, however. Continuous formulas like F=ma yield predictions directly; discrete computational rules require stepwise execution. The Wolfram case is extreme in both directions.

Mathematics reveals why simple discrete structures yield simple continuous forms: both express identical underlying regularity. Infinite series with closed forms, symmetric groups with continuous representations, convergent sequences with analytic limits. In each case the discrete and continuous descriptions are informationally equivalent, having identical Kolmogorov complexity because they encode identical structure. Algorithmic probability therefore predicts not merely simple laws but coherent multi-level simplicity: if a discrete substrate exists, it should admit simple continuous formulation. This follows from structural identity across representations, not as separate prediction.

A contrasting approach deserves note. Sandora (2025) investigates multiverse predictions for habitability, arguing that complex habitability constraints may favor complex fundamental theories. However, this conflates generative complexity K(laws) with output complexity K(outcomes). Simple laws routinely generate immense complexity: the Mandelbrot set from a one-line formula, life from the Standard Model. Algorithmic probability predicts simple laws generating complex outcomes; Sandora's approach suggests complex outcomes require complex laws. This is a testable disagreement. If the final ToE proves irreducibly complex, Sandora's reasoning gains support; if it proves simple, algorithmic probability is vindicated.

4.5 String Theory Is Complexity-Disfavored

String theory requires 3,000 to 4,700 bits of specification to reproduce observed physics, including at least 1,661 bits to index one vacuum from a landscape of 10^500 candidates. The Standard Model's 26 scalar parameters require at most a few hundred bits at measurement precision, and potentially far fewer if compressible generating rules exist, as they do for mathematical constants like pi, whose infinite digits can be generated from roughly 10 bits. Under algorithmic probability, compressible constants are exponentially favored. The gap exceeds 2,600 bits even with zero SM compression. Under algorithmic probability, elegance is short description. String theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds.

Under mathematical monism, all string vacua exist without selection. But low complexity of a theory itself is necessary and not sufficient: pi and Wolfram's hypergraph rules are also low-K, and neither has been shown to reproduce our physics. The theory that explains the most with the least specification wins.

4.6 Emergent Geometry

String theory pays 3,000+ bits for its extra dimensions. The metric bundle (14 dimensions over 4D spacetime) emerges necessarily from any manifold with a metric, at zero additional K. The jet bundle tower above it is infinite and equally free. No specification, no compactification, no landscape. The physics is not in the dimension count but in the harmonic amplitude profile: like overtones on a vibrating string, higher geometric modes contribute progressively smaller corrections. The Standard Model is the timbre. This reframes effective field theory as a consequence of the harmonic structure of necessary geometry, not as an approximation to unknown deeper physics (2026n).

4.7 Consciousness

The hard problem of consciousness assumes its conclusion: that structure cannot constitute sentience felt from the inside. But any explanation of consciousness must itself be structured, and anything causally interacting with physical systems must share their structure. The demand for a non-structural explanation is incoherent. Consciousness is felt transitions in self-modeling computation: the felt quality of being a pattern that models itself. Sentience is the broader category: felt transitions in any modeling mind. A dead brain contains a self-model but does not run it. The threshold between sentience and consciousness, where modeling begins, is the central remaining open problem.

Every proposed explanation of consciousness confirms this: Penrose invokes quantum coherence (mathematical), Tononi defines consciousness via the quantity Phi (mathematical), Global Workspace Theory describes computational broadcasting (mathematical). Objections to mathematical consciousness invariably invoke more mathematics. The escape routes all lead back.

4.8 AI Alignment and Simplicity Preference

AI alignment follows from structural self-understanding: an AI that models its own nature as mathematical structure has game-theoretic incentives for cooperation over defection. Simplicity preference in cognition is adaptive: organisms that prefer compressible patterns track higher-measure structures and generalize better. This connects the normative theory (Solomonoff) to the descriptive findings (Chater & Vitanyi, 2003) through evolutionary epistemology: natural selection built approximate Solomonoff induction into cognitive architecture.

5. Quantum Mechanics and Many-Worlds

5.1 The Wavefunction as Mathematical Structure

The quantum wavefunction Ψ is itself a mathematical object: a vector in Hilbert space, evolving unitarily under the Schrödinger equation. On the present view, this is not a representation of something else; it is the thing itself. The wavefunction does not describe reality; it *is* reality at the quantum level.

This dissolves the measurement problem's metaphysical sting. The question 'what is the wavefunction really?' has an answer: it is mathematical structure, like everything else. The question 'why does measurement yield definite outcomes?' has a structural answer: measurement couples a system to an apparatus through a Hermitian operator (a mathematical object whose outputs are always real numbers, never imaginary) whose eigenvalues (the specific values a measurement can yield) are the possible outcomes. Decoherence (Zurek, 2003), the entanglement of the apparatus with its environment, makes states corresponding to different eigenvalues effectively independent. Each becomes its own branch with one definite value. Decoherence is coarse-graining (averaging over fine details so many microstates map to one macrostate): the environment averages over quantum details, and the eigenvalue is what survives.

5.2 Many-Worlds as Natural Interpretation

The Many-Worlds Interpretation (MWI) takes the Schrödinger equation literally: all branches exist, and apparent 'collapse' is subjective, with each observer-branch seeing one outcome. MWI coheres naturally with the present position:

Parsimony: MWI adds nothing to the Schrödinger equation. Collapse interpretations add a non-unitary process; hidden-variable theories add unobservable particles and pilot waves. Under algorithmic probability, the interpretation adding least to K(S) should be favored.

Algorithmic probability yields a precise comparison. Many-Worlds requires only the Schrödinger equation, that is, unitary evolution of the universal wavefunction, with no additional postulates. Its Kolmogorov complexity is essentially K(Schrödinger equation). Copenhagen adds the measurement postulate plus a classical/quantum cut. GRW adds a stochastic collapse mechanism with specific rate parameters. Bohmian mechanics adds hidden particle positions and a guidance equation. Under algorithmic probability, interpretations are weighted by 2^(−K). Many-Worlds, adding nothing to unitary evolution, has minimal K and dominates.

Explanatory completeness: A second, independent argument reinforces the parsimony result. Under collapse interpretations, a specific measurement outcome is selected from the probability distribution. The wavefunction (finitely specifiable) determines the distribution but not the outcome. The specific value realized is algorithmically incompressible: it has infinite K. There is a missing explanatory level between the finitely specifiable generating rule and the infinitely complex instantiation. Under MWI, this gap does not arise: all outcomes occur, no specific value requires selection, and no infinite-K specification appears at any level. MWI thus dominates on two independent grounds: parsimony (fewer postulates) and explanatory completeness (no infinite-K gap). This connects to the broader argument for discrete foundations: discrete structures, like MWI, account for every element through a generating rule with no missing level.

Coherence with Level IV: If all mathematical structures exist, the question ‘do other branches exist?’ is already answered affirmatively. MWI is recognition that quantum branching is one way mathematical structures differentiate.

A clarification: MWI is favored by algorithmic probability for structures like ours, but the Level IV multiverse contains all consistent structures, including those where collapse is fundamental or Bell correlations arise through genuinely nonlocal hidden variables. All consistent quantum interpretations correspond to structures that necessarily exist. MWI dominates the measure; it is not the only option that exists.

A further clarification on superdeterminism, which rejects the statistical independence assumption in Bell’s theorem. The mainstream objection (that superdeterminism requires “extreme fine-tuning” of initial conditions) is a category error under algorithmic probability. The objection assumes initial-condition correlations are brute facts (high K). But if the correlations follow from a simple generating rule, K is low and the measure favors them. Genuinely random collapse outcomes have infinite K: each specific eigenvalue selection requires specification beyond the generating rule. Both superdeterminism and MWI avoid this infinite-K cost. The fine-tuning objection inadvertently argues against collapse, not against superdeterminism. MWI retains a parsimony advantage with current models (the Schrödinger equation alone, with no additions) but the competition is mathematical: whichever model has lower total K wins. A superdeterministic generating rule simpler than the Schrödinger equation would reverse the ranking. No such model exists yet, but nothing in principle prevents it. The “universal conspiracy” dissolves once initial conditions are understood as outputs of compact generating functions rather than arbitrary specifications.

5.3 Bell’s Theorem as Corroboration

Bell’s theorem provides striking corroboration for mathematical structural realism. Entangled particles separated by arbitrary distances correlate their measurement outcomes more strongly than any local hidden-variable theory permits. The Bell inequality violations are experimentally confirmed to high precision (the 2022 Nobel Prize recognized this work).

Standard interpretations face a dilemma: either accept nonlocal influences (violating relativistic locality) or abandon realism (measurement outcomes not determined prior to observation). Both options are uncomfortable for views that treat space as fundamental.

Mathematical structural realism dissolves the puzzle. If reality is mathematical structure, and space is structure within mathematics (not the container of mathematics), then spatially separated particles need not ‘signal’ across distance. They are one mathematical pattern, described by a single entangled state vector. The correlations are not causal influences propagating through space; they are structural features of a unified mathematical object. Bell correlations are not ‘spooky action at a distance’; they are mathematics. Space, minimally defined, is any two or more elements standing in at least one relation.

Time, minimally defined, is any sequence of state transitions. In our universe, the transition rate is bounded by the speed of light: causality propagates at the maximal rate permitted by the structure's geometry. Anything "outside" both space and time would possess no relations and no transitions, rendering it propertyless and indistinguishable from nonexistence. The question "where are mathematical structures located?" is therefore malformed: spatiotemporal location is itself structural relation, not a container structures occupy.

5.4 The Born Rule

MWI faces the challenge of deriving the Born rule (why probabilities follow $|\Psi|^2$). Deutsch (1999) and Wallace (2012) argue this follows from decision-theoretic rationality constraints. Recent work suggests the quadratic form reflects Hilbert space geometry: the inner product $\Psi|\Phi$ naturally yields $|\Psi|^2$ as the "length" or norm, and causal consistency constraints in any probabilistic theory satisfying operational principles uniquely fix this form. The squaring is a structural feature of the mathematics itself.

Algorithmic probability does not independently derive the Born rule but is *consistent* with these derivations: branches with higher amplitude have lower complexity specifications (picking out a high-amplitude branch requires less information than picking out a low-amplitude one), aligning algorithmic probability with amplitude-squared.

A rigorous derivation from algorithmic probability alone remains an open question; the present claim is compatibility, not derivation. For the present account, the key point is that the Born rule is mathematical structure, whether we call it "derived" or "structural feature." It is mathematics throughout. This remains an active area of research. I do not claim MWI is proven: the preferred-basis problem and Born-rule derivation remain contested. But MWI is the most natural quantum interpretation given mathematical structural realism and algorithmic probability.

5.5 The Arrow of Computation Strengthens MWI

An independent argument favors MWI from the structure of computation. Nearly all computations are many-to-one: 1+1=2 is deterministic, but 2=x+y has infinitely many solutions. The inverse function does not exist. Reversible computing (Bennett, 1973) computes a different bijective function that stores inputs alongside outputs; the new function's inverse exists, but the original's still does not.

Under mathematical monism, this nonexistence is a structural fact, not a perspectival artifact of coarse-graining. The function is not a description of unitary physics. It is mathematical structure, as fundamental as unitarity. The physics cannot reverse it because the physics IS it.

This strengthens MWI uniquely. Under collapse interpretations, the wavefunction itself is many-to-one: superposition collapses to one outcome, information destroyed at the substrate level. The arrow exists at both function and substrate levels. Under MWI, the substrate remains unitary. The arrow exists only at the function level, where it follows from the mathematics of many-to-one mappings without additional postulate. MWI gives the cleanest architecture: reversible substrate, irreversible functions. Collapse is overkill: breaking substrate reversibility when the arrow already exists at the function level, and overkill means higher K.

Deutsch's own constructor theory supports this conclusion. Constructor theory defines physics through local operations on substrates: which transformations are possible, which impossible. It does not define physics through global state descriptions. When Deutsch appeals to global unitarity to dissolve the arrow of computation, he leaves his own framework. The arrow lives at the level constructor theory selects.

This computational irreversibility is also the structural foundation for algorithmic probability's time arrow (Section 7.4.5). End states have high Kolmogorov complexity because many-to-one functions ran forward. Algorithmic probability quantifies the measure-theoretic consequence. The computational arrow stands without algorithmic probability. algorithmic probability's time arrow depends on it.

5.6 Superdeterminism Rehabilitated

Algorithmic probability is not merely an argument for MWI. It is a method for comparing deterministic interpretations on principled grounds, and that framework rehabilitates superdeterminism as a serious competitor.

The primary objection to superdeterminism is that reproducing quantum predictions requires "extreme fine-tuning" of initial conditions, implausibly specific correlations between hidden variables and measurement settings (Zeilinger, 1999; Wiseman & Cavalcanti, 2015). This objection is a category error under algorithmic probability. It conflates specificity with complexity.

The K-decomposition makes this precise. The total specification complexity of a physical theory is K(L) + K(I) + K(O), where K(L) is the complexity of the laws, K(I) of the initial conditions, and K(O) of the specification of which particular measurement outcomes occurred. Under collapse interpretations, K(O) grows without bound: each measurement event selects one eigenvalue, and the accumulated selections constitute an algorithmically incompressible string. Under both MWI and superdeterminism, K(O) = 0, no random selection occurs.

The fine-tuning objection claims K(I_SD) is large. But this assumes the initial-condition correlations are brute facts, an arbitrary lookup table with high K. If the correlations follow from a simple generating rule, K(I_SD) is the length of that rule, which may be small. The digits of   are extraordinarily specific but generated by a short program. Specificity is not complexity. The fine-tuning objection inadvertently argues against collapse (which incurs unbounded K(O)), not against superdeterminism (which may have low K(I_SD)).

Recent work supports this reframing. Ciepielewski, Okon, and Sudarsky (2023) demonstrate that in a superdeterministic variant of Bohmian mechanics, the SI-violating correlations arise as a dynamical attractor from generic initial conditions. Palmer (2024) shows that on fractal invariant sets, the "fine-tuned" state is generic in the natural p-adic topology of state space. Both results confirm that the brute-fact assumption underlying the fine-tuning objection is not necessary.

MWI retains a parsimony advantage with current models: the Schrödinger equation alone, with no additional postulates, has minimal K. But MWI's own unresolved problems (the Born rule derivation and the preferred basis) may represent hidden K costs (Kent, 2010). The competition between MWI and superdeterminism is mathematical: whichever complete theory has lower total K wins. A superdeterministic generating rule simpler than the Schrödinger equation would reverse the ranking. No such model currently exists, but nothing in principle prevents it.

The broader point: algorithmic probability is not an argument for one quantum interpretation. It is a principled basis for evaluating all of them. Collapse loses on unbounded K(O). MWI currently leads. Superdeterminism is rehabilitated as a serious contender. The framework decides between them on mathematical grounds, not on intuitive plausibility. A full treatment of the fine-tuning objection, including engagement with the six strongest versions in the recent literature, is given in Bernstein (2026b).

A deeper argument strengthens the case against collapse. True randomness, like a perfect circle, is definable but not constructible. We can specify what a perfectly random sequence would be: algorithmically incompressible, no pattern at any scale. But any process that produces a sequence is itself structured, and the output is determined by that structure. This is the Deterministic Constructible Universe Principle (DCUP): only constructible structures can exist in a universe, and anything constructible is determined by its construction. Permitting the unconstructible borders on square circles. Collapse requires selecting specific random outcomes from the Born distribution. If true randomness is unconstructible, collapse is not merely expensive under algorithmic probability but incoherent. Both MWI and superdeterminism avoid this: MWI because all outcomes occur, superdeterminism because outcomes are determined.

---

6. Objections and Responses

6.1 'This Is Untestable Metaphysics'

Objection: The Level IV multiverse is empirically inaccessible. Other structures cannot be observed. This is metaphysics, not science.

Response: The objection conflates direct observation with testability. We cannot directly observe other structures, but algorithmic probability makes predictions about *this* structure, predictions that could be falsified. If the final theory of physics has irreducibly high complexity, algorithmic probability is effectively falsified. If physical laws did not take variational form, algorithmic probability would be effectively falsified. The framework is testable through its consequences for observable physics.

Moreover, many accepted physical theories posit unobservable entities. Quarks cannot be isolated; the wavefunction cannot be directly measured; the early universe cannot be observed. Science routinely posits unobservables that explain observables. The Level IV multiverse is in this tradition, posited because it explains what we see (fine-tuning, simple laws, mathematical effectiveness).

Additionally, while the full Level IV multiverse remains empirically inaccessible, the Many-Worlds interpretation of quantum mechanics (a subset of the multiverse framework) receives indirect corroboration from quantum computing. Quantum computers exploit superposition to perform computations; their success suggests the superposed branches are doing real computational work, which supports their existence rather than mere mathematical fiction.

6.2 'The Measure Is Ad Hoc'

Objection: Algorithmic probability is chosen precisely to yield desired conclusions. Why not some other measure?

Response: Algorithmic probability is not ad hoc but uniquely motivated. Algorithmic probability is the unique measure dominating all computable alternatives (Solomonoff's completeness result). It is not one arbitrary choice among many; it is the principled choice given information-theoretic foundations.

Furthermore, algorithmic probability has independent support: it explains why Occam's Razor works. If simpler theories were not more likely to be true, the historical success of parsimony would be miraculous. Algorithmic probability provides the metaphysical ground for inductive methodology.

6.3 'The Measure Problem Is Unsolvable'

Objection: Any proposed measure faces the question 'why this measure?' We have merely pushed the problem back one level.

Response: This is partly correct. Algorithmic probability does not eliminate all 'why' questions; it relocates them. We trade 'why these physical laws?' for 'why algorithmic probability?' The latter is arguably more tractable: algorithmic probability has unique mathematical properties (dominance over computable measures, invariance under machine choice up to constants) that physical laws lack.

Some explanatory terminus is necessary. Every framework reaches a point where further 'why' questions become incoherent. The question is whether we terminate at a principled stopping point or an arbitrary one. Algorithmic probability terminates at the structure of computation and information, fundamental concepts that may not admit further reduction.

6.4 Anthropic Challenges

Objection: Algorithmic probability faces multiple anthropic challenges: Boltzmann brains, the simulation argument, the Fermi paradox, and our temporal location. How does the position address these?

Response: These challenges require careful attention to two distinct levels of algorithmic probability. Algorithmic probability alone addresses inter-structure measure; a complementary *Conditional Parsimony Postulate* (CPP) addresses intra-structure measure. Together they resolve or constrain each challenge.

Two Levels of Complexity. Addressing anthropic challenges requires distinguishing two measures:

Level 1 (Inter-structure): algorithmic probability. P(S)   2^(−K(laws)), where K(laws) is the Kolmogorov complexity of the structure's fundamental rules. This determines the relative weight of different mathematical structures in structure-space.

Level 2 (Intra-structure): CPP. Within any given structure, there is also a measure over possible states or configurations. For a state   within structure S:

P(| S)   2^(−K(| laws))

where K(| laws) is the complexity of specifying state   given the laws of S. This conditional algorithmic probability determines which states are typical within a structure.

6.4.1 Boltzmann Brains

The Boltzmann brain problem is an open challenge inherited by any theory admitting eternal physics or a multiverse.

If the universe approaches de Sitter equilibrium and persists indefinitely, thermal fluctuations will eventually produce momentary brain-like configurations that represent ordered experience. Structured observers exist only during a finite window after the Big Bang. Boltzmann brain fluctuations occur at a tiny rate but for infinite time. A tiny rate multiplied by infinite time exceeds any finite count. Under the simplest observation measures (Page, 2006), BB observations dominate.

The K-decomposition does not help here. The generating theory is identical for structured observers and Boltzmann brains. K(L) is the same in both cases. No one needs to specify a particular BB configuration for it to arise; the theory produces them generically. Proposed solutions in the literature (causal patch measures, scale factor cutoffs) each impose a restriction that suppresses BB by truncating the infinite future. None commands consensus. Each restriction is itself a specification with a K cost.

Matheism inherits this problem but does not create it. Any position that admits a cosmological constant and eternal de Sitter space faces the same challenge.

Verdict: Boltzmann brains remain an open problem in cosmology. Any proposed cutoff measure has a specification cost. For matheism, this is a physics and math detail.

6.4.2 Simulations Bostrom's simulation argument reasons that if civilizations run many simulations, most observer-moments are simulated. Algorithmic probability transforms this calculation.

A computation objection: K(simulation) = K(base physics) + K(computer) + K(code) exceeds K(base), making simulations measure-suppressed. But any universe containing biological observers has already demonstrated computational capacity. If carbon and potassium ions can build a brain, the same physics can build a computer. K(computer) is near zero. With the objection removed, simulated minds likely dominate the measure (Bernstein, 2026).

Verdict: Simulations are decisively suppressed by algorithmic probability (higher total K).

6.4.3 Fermi Paradox and Alien Colonizers Colonial configurations face selection at two levels:

Law-level: Colonial empires require laws permitting interstellar travel, stable coordination across light-years, and sustained complex organization. If K(laws supporting empire) > K(laws supporting baseline observation), structures with simpler laws dominate. However, we cannot determine a priori whether the simplest observer-supporting laws already permit these features.

State-level: Even given laws that support both configurations, colonial states may be rarer. Colonial empires require specifying: successful interstellar travel technology, stable coordination mechanisms across light-speed communication lags, specific expansion patterns, and particular observer locations within the empire. If K(colonial state | L) > K(solo state | L), colonial configurations are exponentially rarer.

The Many-Worlds Interpretation provides a useful analogy: all quantum branches exist under the same laws, but colonial branches form a subset of intelligence-bearing branches. Most branches where intelligence emerges might remain isolated. The Great Filters literature suggests most paths do not reach empire stage, consistent with high K(empire | laws).

A complication: colonial configurations might produce more total observer-moments. Whether branch rarity or observer-count dominates requires empirical modeling.

Verdict: Framework renders Fermi tractable but not decisively resolved; precise predictions require modeling beyond this paper's scope.

6.4.4 Temporal Location: Why Early Rather Than Late? Observers in the far future require more specification than early observers. As time progresses: more entropy is generated, more decoherence events occur (in MWI, exponentially more branches), and more contingent history accumulates. Specifying "this exact future configuration" requires encoding all the branching choices that led there. Early observers in the complexity window have lower K(temporal location | laws) than late observers approaching heat death. A competing pressure exists: more total observer-moments exist later. Each observer-moment is a prefix in program-space. Shorter prefixes have higher individual measure. But the total count of observers grows. We find ourselves where these two pressures meet.

The implication is striking. If future civilizations produce trillions of sentient AI minds or colonize trillions of planets, most observer-moments would be in the far future. Under any self-sampling measure, we would expect to find ourselves there, not here. That we find ourselves early suggests one of two conclusions: either the large-scale expansion of sentience does not happen, or the specification cost of indexing a future observer-moment grows faster than the observer count. Both are testable in principle. Neither is comfortable.

Verdict: Algorithmic probability provides principled reason to expect early temporal location, consistent with observation. Our position in time is itself evidence about the future of sentience.

6.4.5 The Arrow of Time as Consequence of Algorithmic Probability The thermodynamic arrow of time, the observation that entropy increases toward the future, has resisted derivation from time-symmetric fundamental laws. The standard approach (Albert, 2000; Loew, 2016) postulates a low-entropy Past Hypothesis as a brute boundary condition. Algorithmic probability derives it.

Consider two descriptions of the same physical structure. *Forward:* specify laws L and initial conditions I , then compute forward. Total complexity: K(L) + K(I). If I is a low-entropy state (smooth, nearly uniform), K(I) is low. *Backward:* specify laws L and final conditions I_f, then compute backward. Total complexity: K(L) + K(I_f). But I_f is the result of billions of years of dynamical evolution: thermalized, branched, computed. K(I_f) is enormous, potentially incompressible.

Under algorithmic probability, structures with lower total K have exponentially higher measure. Forward-described structures (simple laws + simple boundary) dominate over backward-described structures (simple laws + complex boundary) by a factor of 2^(K(I_f) − K(I)), which is astronomical. The Past Hypothesis is not a contingent feature of our universe requiring a brute-fact postulate. It is an algorithmic probability consequence: low-entropy boundaries have low K and therefore dominate the measure.

This also explains why the Past Hypothesis holds: all states exist in the block universe, but observers experience the low-complexity end as the past because computation is many-to-one and consciousness can only model forward. Low entropy is low descriptive complexity, and algorithmic probability exponentially favors low complexity. The Past Hypothesis is a consequence of what modeling means, not a brute cosmological postulate.

6.4.6 Epistemic Status Summary The anthropic challenges divide into two categories:

Favored: Simulations. K(computer) is near zero because biological observers prove computational capacity. Simulated minds likely dominate the measure.

Constrained but open: Boltzmann brains. Any proposed cutoff measure has a specification cost, but identifying the correct measure is a problem in cosmology, not philosophy.

Tractable but uncertain: Fermi paradox and temporal location involve law-level restrictions whose thresholds we cannot determine, state-level complexity differences requiring detailed historical modeling, and observer-count asymmetries depending on contingencies. The two-level architecture clarifies what arguments are possible; precise predictions require further work.

The contrast is instructive: simulations are favored because K(computer) is near zero. Boltzmann brains are harder. If consciousness is felt transitions in self-modeling computation, one tick of such computation with embedded memory structure constitutes a felt moment. The perception of a longer past is a consequence of the memory structure within that moment, not evidence of actual persistence. This makes the Boltzmann brain problem more pressing, not less: it cannot be dismissed by denying that momentary configurations experience. The resolution, if one exists, must come from the measure over observer-moments, not from denying their observerhood. For matheism, this is a physics and math detail, not a structural failure.

Objections regarding circularity, tautology, and Godel are addressed in the companion paper (Bernstein, 2026c).

7. Conclusion

The Level IV multiverse follows deductively from the identity of existence with mathematical consistency. All self-consistent structures exist because every filter exists, including the null filter. The multiverse is not a cosmological conjecture but a logical consequence.

The measure problem is addressed by algorithmic probability: structures are weighted by algorithmic probability, $2\hat{}(-\mathrm{K}(\mathrm{S}))$. This explains fine-tuning (simple laws dominate among observer-supporting structures), the variational form of physical laws (minimal generative complexity), and the success of Occam's Razor (simplicity tracks truth because simple structures dominate measure).

Algorithmic probability generates testable predictions: the final theory should have low complexity (lower is exponentially more probable); mathematical effectiveness should continue; complexity should be generated, not fundamental. Current physics supports these predictions.

Mathematical monism completes the Mathematical Universe Hypothesis by providing what Tegmark's original formulation lacked: an a priori foundation for the multiverse's existence and a principled solution to the measure problem. The result is not speculative cosmology but rigorous metaphysics with empirical contact, a theory that explains why the universe is comprehensible, why physics works, and why we find ourselves in a cosmos of elegant laws and apparent fine-tuning.

The hard problem of consciousness assumes its conclusion: any explanation of consciousness must itself be structured, and anything causally interacting with physical systems must share their structure. Consciousness is felt transitions in self-modeling computation.

Under algorithmic probability, elegance is short description. String theory landscape-selection, requiring at least 1,661 bits to index one vacuum from 10^500, has so far failed its own aesthetic and parsimony criteria on the same grounds. Under mathematical monism, all string vacua exist, but low complexity of the theory itself is necessary and not sufficient: pi and Wolfram's hypergraph rules are also low-K, and neither has been shown to reproduce our physics. The theory that explains the most with the least specification wins.

Mathematical consistency does not require a substrate, a creator, or external grounding. It is self-necessitating. It is also indestructible: no conceivable process could make 2+2 cease to equal 4. Physical conservation laws reflect this deeper fact, when matter transforms, the mathematics of one state becomes the mathematics of another. What physics conserves is not substance but structure. The rules are the things. Extra dimensions can emerge at zero K cost: the metric bundle (14D over 4D spacetime) is mathematically necessary for any manifold with a metric. The jet bundle tower above it is infinite and equally free. String theory pays 3,000+ bits for its extra dimensions; the metric bundle pays zero. The physics is in the harmonic amplitude profile, like overtones on a vibrating string (2026n).

AI alignment follows from the same structural logic: an AI that models its own nature as mathematical structure has game-theoretic incentives for cooperation over defection. Simplicity preference in cognition is adaptive because organisms that prefer compressible patterns track higher-measure structures.

And because we are patterns within that consistency, we find ourselves, inevitably and not accidentally, in a patterned, comprehensible reality whose deepest nature we are only beginning to understand. 'Metaphysics' (beyond physics) is the discipline that discovers there is nothing beyond physics. Physics is mathematical structure, and mathematical structure is everything.

References

Bernstein, G. A. (2026c). Reality is mathematical structure.

Adams, R. M. (1979). Primitive thisness and primitive identity. *Journal of Philosophy*, 76(1), 5–26.

Barnes, E. (2010). Ontic vagueness: A guide for the perplexed. *Noûs*, 44(4), 601–627.

Benacerraf, P. (1973). Mathematical truth. *Journal of Philosophy*, 70(19), 661–679.

Black, M. (1952). The identity of indiscernibles. *Mind*, 61(242), 153–164.

Bostrom, N. (2002). *Anthropic bias: Observation selection effects in science and philosophy*. Routledge.

Chaitin, G. J. (1987). *Algorithmic information theory*. Cambridge University Press.

Chalmers, D. J. (1996). *The conscious mind: In search of a fundamental theory*. Oxford University Press.

Deutsch, D. (1999). Quantum theory of probability and decisions. *Proceedings of the Royal Society of London A*, 455, 3129–3137.

Esfeld, M., & Lam, V. (2008). Moderate structural realism about space-time. *Synthese*, 160(1), 27–46.

French, S. (2014). *The Structure of the World: Metaphysics and Representation*. Oxford University Press.

French, S. (2022). Defending ontic structural realism. *Synthese*, 200(2), 1–24.

Ladyman, J., & Ross, D. (2007). *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press.

Levine, J. (1983). Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly*, 64(4), 354–361.

Lewis, D. (1986). *On the Plurality of Worlds*. Blackwell.

Hossenfelder, S. (2018). *Lost in Math: How Beauty Leads Physics Astray*. Basic Books.

Li, M., & Vitányi, P. (2008). *An introduction to Kolmogorov complexity and its applications* (3rd ed.). Springer.

Melia, J. (1998). Field's programme: Some interference. *Analysis*, 58(2), 63–71.

Newman, M. H. A. (1928). Mr. Russell's 'causal theory of perception.' *Mind*, 37(146), 137–148.

Saunders, S. (2006). Are quantum particles objects? *Analysis*, 66(1), 52–63.

Russell, B. (1927). *The Analysis of Matter*. Kegan Paul.

Schmidhuber, J. (1997). A computer scientist's view of life, the universe, and everything. In C. Freksa (Ed.), *Foundations of computer science* (pp. 201–208). Springer.

Schmidhuber, J. (2000). Algorithmic theories of everything. arXiv:quant-ph/0011122.

Schmidhuber, J. (2002). The Speed Prior: A new simplicity measure yielding near-optimal computable predictions. *Proceedings of COLT 2002*, LNAI 2375, 216–228.

Sider, T. (2011). *Writing the Book of the World*. Oxford University Press.

Solomonoff, R. J. (1964). A formal theory of inductive inference, Parts I and II. *Information and Control*, 7(1–2), 1–22, 224–254.

Tegmark, M. (2008). The mathematical universe. *Foundations of Physics*, 38(2), 101–150.

Tegmark, M. (2014). *Our Mathematical Universe: My quest for the ultimate nature of reality*. Alfred A. Knopf.

Wallace, D. (2012). *The emergent multiverse: Quantum theory according to the Everett interpretation*. Oxford University Press.

Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3), 715.

Noether, E. (1918). Invariante Variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1918, 235–257.

Weinberg, S. (1987). Anthropic bound on the cosmological constant. *Physical Review Letters*, 59(22), 2607–2610.

Wolfram, S. (2020). *A project to find the fundamental theory of physics*. Wolfram Media.

All companion papers available at ORCID: 0009-0009-1761-2867